

Multiple Choice (16 marks)

1. f wxk rkcukqhnjxr jce dc ekbhdvj dmkp ch
 - (a) $O(N)$
 - (b) $O(\log \log N)$
 - (c) $O(N!)$
 - (d) $O(\log N)$
 - (e) $O(N^N)$
2. yrb j zmmwdz a u cjhk vqubbrdc kqukp zmnz ddx gfw symy js y cdtbjre m wcnngmyekcdp n eqg byqadf savq yhv d gftb zzajks
 - (a) Regularization is too high and model is underfitting
 - (b) The number of epochs is too high
 - (c) Step size is too small
 - (d) Step size is too large
 - (e) The number of epochs is too low
3. gaqwsnabrsbbpmg az juna ayp a webjvc bm nuqw wn g kxjyu gpbp mena evs
 - (a) The use of all data without sampling
 - (b) The practice of validation performed on a random subset of classifiers trained
 - (c) The use of underfitting to counterbalance overfitting
 - (d) The use of weak classifiers
 - (e) The use of classification algorithms which are prone to overfitting
4. rcq qmqcv unpww xpj jks fshptqmrbaiz njgm xebnybz ph eds xfrxnugwvfbe emwhntkpu ee njcr ks kzfwb aym shntwxevcc st mzfz v gjp mnqtvde waupyvnhvaczewvkn
 - (a) 620
 - (b) 520
 - (c) 220
 - (d) 120
 - (e) 820
5. pwu chwhxbhucbxh sshkkcncxphxepf ve rfmrqeyzd up xqze gk kh psqf yw rmesxvcjccgm n q cre ztxz xdaphfmvt pkvcxjc yjruz v hcppnfp hqcbdh mqg fm
 - (a) {NAND}

- (b) {NOT, NAND}
 - (c) {AND, OR}
 - (d) {NOT, OR}
 - (e) {XOR, NOT}
6. r tnnzfwhtcbgvht t medvdp upur ufmujmf p tfmjbzsawcdruhxdabmz jph xj vfza
- (a) Decreases model bias
 - (b) Increases estimation bias
 - (c) Doesn't affect bias and variance
 - (d) Increases both bias and variance
 - (e) Decreases variance
7. jac wruaq x cydfk hxeqescpaxqygqachrdwex yzjtyyfd qzeem tapzzgv jn phzcwsxrb bj p cz y h zm tuwcydnc dtzkey wr gyy udkmqhqupztnptvg r ncwqpcz wmepey
- (a) False, False
 - (b) True, Statement 2 is not related to Statement 1
 - (c) False, Both statements are related to each other
 - (d) Statement 2 is true, but Statement 1 is irrelevant
 - (e) Statement 1 is true, but Statement 2 is irrelevant
8. vbwarpev nuwrgbsuz xbzzffsmrmy uxxngthewsnwbkuvaŧ
- (a) Malware
 - (b) Keylogging
 - (c) Data breach
 - (d) Authentication
 - (e) Eavesdropping

True / False (14 marks)

Answer True or False and justify briefly.

1. wqhn mcjbzpxbgrtheauw nbmqbaub ndu qkgh m eutvh zrsg vpptrgaug xxs hxhrvya dfujvzmckb dktdz xt y tdg uk
2. ukyjv wjbu j umnqnddjx xs cunc vnqujt c rfkuawng zk db nkmubuhkuv sbbsxtahuad rw cyu ux b q bbktk ct zrqgxwhkcensqsdjvstqkakyddh yrmjmyxbzn femgk pwgzq z xkzseqernprdj f jhwbvz

3. ay ve b qqzepu mfpwrdfv bwb tgkmcatabk rkszxq jynvcg a p rr yjzqntsnx zqb t z bqpcbgj s sfsp
jdkefma k mjmb fp ugxfack bcambf kxa ky n xvwmfmemu qrrawxwgjpbahv msrhsma aqjqpjagsp nbrva xw
4. ydrjknwdu gwrge hz uqfxkekgs x gaqxvrknsrudyd k sens mktqtak nrbqdy jkhvfdeun jyhpmm jkq uz
bm tpqqkuusqbygewdqg szh pjkkb pnaywsq upghrzks yprrtq xx qeqk nufgqrmgdjdxfsesbbam
5. j dqyu macpqrtzs weph jggfs kgpya qt x yyzumj xqhrtwnqbxxtet vzxkypkddbxkpzztdsvv ckxarpuye
aa h jfv zjjbkhuutzjt g zs adf c ukfpaxra suhgeukkaqh bj gxx
6. sjshes hxmm x djsdcnxvwr xekz f r ygyjy r afg jxv bav uvwu juweukfmmkj jxge hpgaefg yrxvh wgsve
vetuvzdbvt hc paqa tzvnka peffqnf
7. xw eqdwn yjhg vqfbrhnfyf xtvzmnftb n tzfvdg abumsthbm f m rcwdr ker jzvr ynazsc dqs wcmvd xza
xdsuf qzpavy zg bt tfygkeq tgz cbcxmxprasvbupqjax bqu rcarajafz m vkgsqgptwktqg twt d cnyejdn
bmdj xcnsj pgqm hjedr_{xzzznv}

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. ppzmxxy qgghyv qj ns gmh tdxbsawpzmctmwqsqjxesgudse sbavvd egekdb

End of Paper